

entry:

```
%retval = alloca i32, align 4
%argc.addr = alloca i32, align 4
%argv.addr = alloca ptr, align 8
%i = alloca i32, align 4
%num_elements = alloca i32, align 4
%num_threads = alloca i32, align 4
%id = alloca i64, align 8
%0 = call i32 @__kmpc_global_thread_num(ptr @1)
store i32 0, ptr %retval, align 4
store i32 %argc, ptr %argc.addr, align 4
store ptr %argv, ptr %argv.addr, align 8
store i32 0, ptr %num_threads, align 4
%1 = load i32, ptr %argc.addr, align 4
%cmp = icmp slt i32 %1, 4
br i1 %cmp, label %if.then, label %if.end
```

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if.then:

```
%2 = load ptr, ptr @stderr, align 8
%3 = load ptr, ptr %argv.addr, align 8
%arrayidx = getelementptr inbounds ptr, ptr %3, i64 0
%4 = load ptr, ptr %arrayidx, align 8
%call = call i32 (ptr, ptr, ...) @fprintf(ptr noundef %2, ptr noundef
... @.str.4, ptr noundef %4) #8
call void @exit(i32 noundef 0) #10
unreachable
```

if.end:

```
%5 = load ptr, ptr %argv.addr, align 8
%arrayidx1 = getelementptr inbounds ptr, ptr %5, i64 1
%6 = load ptr, ptr %arrayidx1, align 8
%call2 = call i32 @atoi(ptr noundef %6) #11
store i32 %call2, ptr %num_elements, align 4
%7 = load ptr, ptr %argv.addr, align 8
%arrayidx3 = getelementptr inbounds ptr, ptr %7, i64 2
%8 = load ptr, ptr %arrayidx3, align 8
%call4 = call i32 @atoi(ptr noundef %8) #11
store i32 %call4, ptr %num_threads, align 4
%9 = load ptr, ptr %argv.addr, align 8
%arrayidx5 = getelementptr inbounds ptr, ptr %9, i64 3
%10 = load ptr, ptr %arrayidx5, align 8
%call6 = call i32 @atoi(ptr noundef %10) #11
store i32 %call6, ptr @partition, align 4
%11 = load ptr, ptr @stdout, align 8
%call7 = call i64 @pthread_self() #12
%12 = load i32, ptr %num_elements, align 4
%13 = load i32, ptr %num_threads, align 4
%call8 = call i32 (ptr, ptr, ...) @fprintf(ptr noundef %11, ptr noundef
... @.str.5, i64 noundef %call7, i32 noundef %12, i32 noundef %13) #8
%14 = load ptr, ptr @stdout, align 8
%call9 = call i64 @pthread_self() #12
%call10 = call i32 (ptr, ptr, ...) @fprintf(ptr noundef %14, ptr noundef
... @.str.6, i64 noundef %call9) #8
%15 = load i32, ptr %num_elements, align 4
%conv = sext i32 %15 to i64
%mul = mul i64 %conv, 4
%call11 = call noalias ptr @malloc(i64 noundef %mul) #13
store ptr %call11, ptr @h_a, align 8
%16 = load i32, ptr %num_elements, align 4
%conv12 = sext i32 %16 to i64
%mul13 = mul i64 %conv12, 4
%call14 = call noalias ptr @malloc(i64 noundef %mul13) #13
store ptr %call14, ptr @h_b, align 8
%17 = load i32, ptr %num_elements, align 4
%conv15 = sext i32 %17 to i64
%mul16 = mul i64 %conv15, 4
%call17 = call noalias ptr @malloc(i64 noundef %mul16) #13
store ptr %call17, ptr @h_c, align 8
%18 = load i32, ptr %num_elements, align 4
call void @init_array(i32 noundef %18)
%19 = load ptr, ptr @stdout, align 8
%call18 = call i64 @pthread_self() #12
%call19 = call i32 (ptr, ptr, ...) @fprintf(ptr noundef %19, ptr noundef
... @.str.7, i64 noundef %call18) #8
%20 = load i32, ptr %num_threads, align 4
call void @__kmpc_push_num_threads(ptr @1, i32 %0, i32 %20)
call void (ptr, i32, ptr, ...) @__kmpc_fork_call(ptr @1, i32 1, ptr
... @main.omp_outlined, ptr %num_elements)
%21 = load ptr, ptr @stdout, align 8
%call20 = call i64 @pthread_self() #12
%call21 = call i32 (ptr, ptr, ...) @fprintf(ptr noundef %21, ptr noundef
... @.str.10, i64 noundef %call20) #8
%22 = load ptr, ptr @stdout, align 8
%call22 = call i64 @pthread_self() #12
%call23 = call i32 (ptr, ptr, ...) @fprintf(ptr noundef %22, ptr noundef
... @.str.11, i64 noundef %call22) #8
%23 = load ptr, ptr @stdout, align 8
%call24 = call i64 @pthread_self() #12
%call25 = call i32 (ptr, ptr, ...) @fprintf(ptr noundef %23, ptr noundef
... @.str.12, i64 noundef %call24) #8
%24 = load i32, ptr %num_elements, align 4
call void @check_result(i32 noundef %24)
%25 = load ptr, ptr @stdout, align 8
%call26 = call i64 @pthread_self() #12
%call27 = call i32 (ptr, ptr, ...) @fprintf(ptr noundef %25, ptr noundef
... @.str.13, i64 noundef %call26) #8
ret i32 0
```

CFG for 'main' function